

END TERM EXAMINATION

SECOND SEMESTER [BCA] JULY 2023

Paper Code: BCA-102

Subject: Applied Mathematics

(Batch 2021 Onwards)

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit. Scientific Calculator is allowed.

Q1 Answer the following:-

- (a) Determine the binomial distribution for which mean is 4 and variance is 3. (2.5)
- (b) The probabilities that students A, B, C and D solve a problem are $1/3$, $2/5$, $1/5$ and $1/4$, respectively. If all of them try to solve the problem, what is the probability that the problem will be solved? (2.5)
- (c) Construct forward difference table from the values of x and y given below: (2.5)

X	0	1	2	3	4	5
y	5	11	22	18	27	2

- (d) Prove that $\Delta \nabla = \Delta - \nabla$, where Δ and ∇ are forward and backward difference operators. (2.5)
- (e) Using Lagrange interpolation, find the unique polynomial of degree 2, such that $f(0) = 1$, $f(1) = 3$, $f(3) = 55$. (2.5)
- (f) What do you mean by Numerical Integration? (2.5)
- (g) Calculate $\int_1^5 \sin x \, dx$ correct to four decimal places with $h = 1$. (2.5)
- (h) Define basic feasible solution in Linear Programming Problem (LPP). (2.5)
- (i) Write a note on unbounded solution of LPP. (2.5)
- (j) Write a note on slack and surplus variables in LPP. (2.5)

UNIT-I

- Q2 (a) An insurance company insured 2000 scooter drivers, 4000 car drivers and 6000 truck drivers. The probability of an accident involving a scooter driver, car driver and truck driver is 0.01, 0.03 and 0.15 respectively. One of the insured person meets with an accident. What is the probability that he is a truck driver? (6.5)

- (b) If X is a Poisson variate such that $P(X=2) = 9P(X=4) + 90P(X=6)$. Find the mean and variance of X . (6)

- Q3 (a) If the heights of 500 students are normally distributed with mean 68 inches and standard deviation 3 inches. How many students have heights (i) greater than 72 inches (ii) less than equal to 64 inches (iii) between 65 and 71 inches? Given $P(Z \leq 1.33) = 0.4082$, $P(Z \leq 1) = 0.3413$. (6.5)
- (b) A continuous random variable X has probability density function. (6)

$$f(x) = \begin{cases} \frac{k}{1+x^2}, & -\infty < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

Determine the value of k and evaluate $P(X \geq 0)$.

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UNIT-II

- Q4 (a) Find the missing value of the following table. Explain why the result differs from 16? (6)

X	1	2	3	4	5	6	7
y	2	4	8	-	32	64	128

- (b) Find a real root of the equation $x^3 - x = 1$ using False Position method upto fifth iteration. (6.5)

- Q5 (a) Obtain a formula to calculate cube root of a natural number N and use it to evaluate cube root of 28. (6)
- (b) Estimate by suitable method of interpolation the number of persons whose daily income is Rs. 19 but not exceed Rs. 25 from the following data. (6.5)

Income (In Thousands)	0-9	9-19	19-28	28-37	37-46
No. of persons	50	70	203	406	304

UNIT-III

- Q6 (a) Find the LU decomposition of the matrix $A = \begin{bmatrix} 3 & 2 & 7 \\ 2 & 3 & 1 \\ 3 & 4 & 1 \end{bmatrix}$. (6.5)
- (b) From the following table of values of x and y, obtain dy/dx at x = 4 and d^2y/dx^2 at x = 5. (6)

x	0	1	2	3	4	5
y	4	8	15	7	6	2

- Q7 (a) Solve the following system using the Gauss - Seidel iterative method. Perform three iterations only. (6.5)

$$\begin{aligned} 5x - y + 3z &= 3 \\ 4x + 7y - 2z &= 2 \\ 6x - 3y + 9z &= 9 \end{aligned}$$

- (b) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's 1/3 rule and hence obtain the approximate value of π . (6)

UNIT-IV

- Q8 Solve the following linear programming problem by using simplex method: (12.5)
- Min $Z = x_1 + 2x_2 + 3x_3$
- subject to
- $$2x_1 - x_2 + x_3 \geq 4$$
- $$x_1 + x_2 + 2x_3 \leq 8$$
- $$x_2 - x_3 \geq 2$$
- and $x_1, x_2, x_3 \geq 0$

- Q9 (a) Find the optimal solution of the following minimization transportation problem: (6.5)

	D1	D2	D3	D4	Supply
S1	11	13	17	14	250
S2	16	18	14	10	300
S3	21	24	13	10	400
Demand	200	225	275	250	

- (b) A computer centre has 4 expert programmers. The centre wants 3 application programmes to be developed. The head of the computer centre, after studying carefully the programmes to be developed, estimates the computer time in minutes required by the experts for the application programmes as follows: (6)

		Application Programmes		
		A	B	C
Programmers	1	15	18	10
	2	14	17	8
	3	16	19	17
	4	20	14	17

How should the application programmes must be allocated, one per programmer, so as to minimize the total time?

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SECOND SEMESTER (BCA) JULY-2023

Paper Code: BCA-104

Subject: Web Based Programming

Time: 3 Hours

Maximum Marks :75

Note: Attempt five questions in all including Q.no.1 which is compulsory.. Select one question from each unit.

- Q1 Attempt **Any ten** of following: (10x2.5=25)
- a) What are the different types of errors in PHP?
 - b) How can we find out the name of the current executing file?
 - c) Why we use \$_REQUEST variable?
 - d) What is the difference b/w isset and empty?
 - e) What are the differences between Print & echo statements?
 - f) What is juggling in PHP?
 - g) What is the variable-naming rules you should follow in PHP?
 - h) Difference between GET, POST and REQUEST method.
 - i) What is Open-Source Software?
 - j) What are the differences between Print & echo statements?
 - k) What is the difference between "&"&"?
 - l) What is a lexical structure of PHP?

UNIT-I

- Q2 a) What is variable & constant in PHP? Explain the scope of variables in PHP with example. (6.5)
b) Explain the Super Global Variables in PHP? (6)
- Q3 a) Difference between include (), include once (), required (), required once () and header('Location'). (6)
b) What are the different types of Arrays? Explain its benefits. (6.5)

UNIT-II

- Q4 a) Explain Session and Write a PHP script to count the number of visits of web page by the user using session. (6.5)
b) What are the differences between Cookies & Sessions? (6)
- Q5 a) Differentiate Call by Value & Call by Reference concept with suitable example. (6.5)
b) Write a PHP code to design Contact Us form & provide validation on email & name. (6)

UNIT-III

- Q6 a) What is File handling in PHP? Write a PHP script to open, close, read & write into a file. (6)
b) Briefly explain the concept of OOPs and how can you say that PHP is an object-oriented language? (6.5)

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- Q7 a) Write a program using PHP to upload a file.
b) Define Classes & Objects. What are the access modifiers?

(6)
(6.5)

UNIT-IV

- Q8 a) What is MYSQL? Explain datatypes of MYSQL. Write the various constraints of SQL and Write the steps to design a table using PhyMyadmin.
b) Write a program to select all the records and display it in table.
- Q9 a) Write a PHP script to create a database employee and create a employee table with in the database with field name i.e. emp_id, emp_name, emp_salary and emp_designation. Insert a record and select all records of employee table and display in tabular format.
b) In how many ways we can retrieve the data in the result set of MySQL using PHP?

(6.5)

(6)

(6)

(6.5)

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SECOND SEMESTER [BCA] JULY 2023

Paper Code: BCA-110

Subject: Environmental Studies

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit.

(5x5=25)

Q1 Write short notes on the following (Any Five):-

- (a) Importance of Environmental Studies.
- (b) Components of environment.
- (c) Sustainability.
- (d) Ecosystem preservation and conservation strategies.
- (e) Acid rain.
- (f) Environmental Management System.

UNIT-I

- Q2 (a) What are the components of environment? Explain by giving suitable illustrations/examples. (6.5)
(b) Discuss the multidisciplinary nature of environmental studies. (6)
- Q3 (a) Discuss the different components of the atmosphere. (6.5)
(b) Briefly discuss the following:- (6)
(i) Climate Change
(ii) Earth Summit

UNIT-II

- Q4 (a) What do you understand by 'eco system'? (6.5)
(b) Elaborate upon the different ecosystem services. (6)
- Q5 (a) What are renewable and non-renewable energy sources. Explain giving suitable examples. (6.5)
(b) Briefly discuss the following:- (6)
(i) Food Chain
(ii) Ecological succession

UNIT-III

- Q6 (a) Explain the term 'Biodiversity'. (6)
(b) Elaborate upon the endemic and endangered species of India by giving suitable examples. (6.5)
- Q7 Discuss the following:- (6.5)
(a) Value of biodiversity (6)
(b) Biodiversity conservation strategies

UNIT-IV

- Q8 (a) Discuss the different types of environmental pollution by citing its mitigation strategies. (8.5)
(b) What are the harmful effects on human health associated with nuclear emissions. (4)
- Q9 (a) Discuss the control measures for various types of wastes. (6.5)
(b) Write a note on the following:- (6)
(i) Air quality index
(ii) Waste segregation

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SECOND SEMESTER [BCA] JULY 2023

Paper Code: BCA-108

Subject: Database Management System

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No. 1 which is compulsory. Select one question from each unit.

- Q1. Answer the following (any five) (5×5=25)
- (a) Explain advantage of database management system over file processing system.
 - (b) Explain the role of E-R model in database design.
 - (c) Differentiate between serializable and non-serializable transactions.
 - (d) What problems are encountered if data is not stored in normalized table?
 - (e) Explain DDL and DML. Give three commands each for DDL and DML.
 - (f) What is data independence? What are two forms of data independence? Explain with the help of an example.
 - (g) What is the difference between strong and weak entity sets? Why sometimes weak entity sets are needed in database design.

UNIT-I

- Q2. a) Describe the three - schema architecture. Why do we need mappings between schema levels? How does different schema definition language support this architecture? (6)
- b) A database is being constructed to keep track of the teams and games of a sports league. A team has a number of players, not all of whom participate in each game. It is desired to keep track of the players participating in each game for each team, the positions they played in that game, and the result of the game. Design an ER diagram for this application, stating any assumptions you make. Choose your favourite sport (e.g., soccer, baseball, football) (6.5)
- Q3. a) Explain different types of keys with example. Differentiate between Primary key, candidate key and super key. (6.5)
- b) Illustrate the use of SUM (), AVG (), COUNT (), MIN (), MAX (). (6)

UNIT -II

- Q4. a) Consider the following relations
- Customer (CustId, CusName, Address, State)
- Parts (PartNum, Description, Price)
- Order (OrderNo, Name, Partnum, Qty, CustId)

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Write query for each of the following: (2x4=8)

- i) Find all customers who have placed order for part description "Tyres".
 - ii) Find customer name, address of customers who have purchased Partnum 10 and quantity ordered is more than 100.
 - iii) Find the customer name, address of those customers residing in state = "Delhi"
 - iv) Find all customer names that have placed order for product with price more than 500.
- b) Explain stored procedure and triggers in DBMS with example. (4.5)

- Q5. a) What are the integrity constraints? Explain each of them. (6.5)
- b) "NULL value concept is useful one but a large use of NULL VALUE in implemented database is not desirable". Comment. (6)

UNIT-III

- Q6. a) Describe Full Join and Division Operation relational algebraic operations giving one an example of each (6.5)
- b) Define the following terms: (1.5x4=6)
- (i) Normalization
 - (ii) MVD
 - (iii) FFD
 - (iv) BCNF

- Q7. a) Explain Codd's rules in RDBMS. (6)
- b) What are set operations? Explain each with the help of an example. When two tables are said to be union compatible? How are the results ordered in union compatible? (6.5)

UNIT -IV

- Q8. a) What is a transaction? Explain ACID properties of a transaction. (3+3=6)
- b) Discuss the different types of transactions failures that may occur in database environment. (6.5)

- Q9 a) Explain 2 phase locking scheme for data recovery. How two-phase Locking helps in maintaining integrity in the database? (6.5)
- b) Explain Discretionary Access Control [Grant/Revoke] methods for database security. (6)

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SECOND SEMESTER [BCA] JULY 2023

Paper Code: BCA-106

Subject: Data Structure and Algorithm Using C

Maximum Marks: 75

Time: 3 Hours

Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit.

(2.5×10=25)

Q1 Answer the following:-

- (a) Define an Algorithm.
- (b) What is Dynamic Memory Allocation method?
- (c) List down any four application of data structure.
- (d) Define Stack and Queue.
- (e) Define Graphs and Tree.
- (f) Define the hash function.
- (g) What are the asymptotic notations?
- (h) Define the Acyclic graph.
- (i) What are Binary Trees?
- (j) Define adjacency matrix.

UNIT-I

- Q2 (a) What is Sparse Matrix and how will you represent Sparse Matrix by 2D Array? (8.5)
- (b) What is Time Complexity also write the Time Complexity of Selection Sort, Bubble Sort, Insertion Sort, Heap Sort, Quick Sort, Merge Sort, Radix sort? (4)

OR

- Q3 (a) Consider the following array: Arr= 14, 33, 27, 35, 10, Sort this array using Bubble sort Algorithm. (9.5)
- (b) Explain in Simple term how Hash Tables are implemented? (3)

UNIT-II

- Q4 (a) What is Dynamic Memory Allocation and how can you determine the size of an allocated portion of memory? (6.5)
- (b) Write the Difference between:
(i) Static and Dynamic Memory Allocation
(ii) Calloc() and Malloc() (6)

OR

- Q5 (a) Write a Program in C to create and Display a Singly Linked List. (6.5)
- (b) Write an algorithm for Binary Search and also write a simple Binary Search Program in C. (6)

UNIT-III

- Q6 (a) Write a Program to Reverse a String using Stack. (6)
- (b) Write the steps to Convert Infix Expression to a Postfix Expression and Convert an Infix Expression $\text{exp} = "a+b*c+d"$ to Postfix Expression. (6.5)

OR

- Q7 (a) Write a Short note on:

- (i) Linear Queue
- (ii) Circular Queue
- (iii) Priority Queue

- (b) What is Abstract Data Types and its features, also write the advantages and Disadvantages of Abstract Data Types. (6.5)

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